

Question	Answer	Marks
6(a)	circle unit, N circle units, N m ⁻² circle 10, 1000	A3
6(b)	The advancement in Science will require better precision at the realm of the very small objects. The physical sample has its associate uncertainty in mass. Any mass measuring instrument calibrated from it will inherit this uncertainty. When that mass measuring instrument is measuring very small masses, the percentage uncertainty in mass will increase with the decreasing mass measured.	A1
6(b)(i)	Using Fleming's left hand rule: left to right	A1
6(b)(ii) 1.	The free electrons in the rod move together with the rod as the rod moves upwards. The electrons experience a magnetic force as they move through the magnetic field. This force causes the electrons to move to one end of the conductor, leaving a net positive charge on the other end, setting up a potential difference between the ends of the conductor. OR As the rod moves upwards, it cuts the magnetic flux or the associated magnetic flux linkages changes. By Faraday's law, an e.m.f. (that is proportional to the rate of cutting of magnetic flux) will be induced across the two ends of the rod.	B1 B1
6(b)(ii) 2.	At equilibrium, the magnetic force <u>on the charge</u> is balanced by the electric force due to the internal electric field set up at the two ends. $F_B = F_E$ $Beu = eE = e \frac{V}{L}$ $V = BLu$ OR $V = \frac{d\Phi}{dt} = \frac{d(BLx)}{dt} = BLu \quad \text{where } x \text{ is the change in position of conductor in time } dt.$	B1 B1
6(b)(iii)	$m = \frac{BIL}{g} \quad [\text{from (c)(i)}]$ $= \frac{VIL}{Lug} \quad [\text{substitute B using (c)(ii)}]$ $= \frac{\frac{hC}{2e} \times \frac{kne}{2}}{ug}$ $= \frac{nCk}{4ug} h$	B1 A1

	Max Marks	11
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Question	Answer	Marks
7a	There is a magnetic flux of 1 weber through a surface	B1